

Effect of Renewables and Other Factors on Fossil Fuel Power Generation in the UK from 2015-2025

1 Overview

The sources of power generation in the UK have changed dramatically over the last decade, with renewable energy increasingly displacing fossil fuels. This report examines the effect of renewable energy sources, demand, net imports, and time of year on fossil fuel power generation in the UK from 2015-2025.

Multiple exploratory visualisations were used to establish context, showing a clear upward trend in renewable energy production and a corresponding decline in fossil fuel energy production. Coal production in particular declined from nearly 75 TWh in 2015 to zero by 2025, while wind energy grew from 23 TWh to 70 TWh over the same period. Finally, a multiple linear regression model with an R^2 of 0.968 was used to quantify the relationship between the predictor variables and fossil fuel production using daily aggregated data. Analysing the displacement coefficients of the predictor variables and their confidence intervals helped determine the effect of each factor on fossil fuel production.

The model indicates that wind energy and net imports have near-identical displacement coefficients, suggesting that they displace fossil fuels at the same rate. Demand is positively associated with fossil fuel production, while solar energy shows a borderline positive coefficient, likely reflecting multicollinearity with demand. December, May, and June show statistically significant reductions in fossil fuel production relative to January, consistent with high wind generation partially compensating for the higher winter demand.

2 Introduction

Context and motivation The UK National Grid is the high-voltage electric power transmission network that serves the majority of Great Britain, ensuring that the electricity generated meets demand across the country in real time [1]. In recent years, the energy mix of the UK grid has undergone a significant transformation, driven by government policy, falling renewable energy costs, and growing environmental concerns.

Recently, 2024 became the first full year where low-carbon renewables generated more electricity than fossil fuels in the UK, with wind power becoming the single largest source of UK electricity [2]. The decline of fossil fuels as an energy source in the UK has raised concerns regarding energy security and employment, particularly regarding renewable energy jobs not being created quickly enough to match the decline in the North Sea oil and gas industry [3]. As a result, understanding the various factors that drive this displacement, and quantifying the extent to which renewables, imports, and seasonal factors affect fossil fuel production, naturally becomes an area of significant practical and policy relevance.

Previous work Several studies have examined the relationship between renewable energy and fossil fuel displacement. Karlilar Pata and Balcilar [4] analysed fossil fuel displacement by renewables across 36 OECD countries from 2000-2020, finding that wind capacity has a statistically significant displacement coefficient of -0.55, while solar has a smaller effect of -0.49. Their findings suggest that while renewables are progressively displacing fossil fuels, a significant transition is still required for OECD countries to achieve one-to-one displacement with variable renewables.

At the UK level, Carbon Brief [5] found that fossil fuels fell to their lowest ever share of UK electricity in 2023 at just 33%, driven by a six-fold expansion of renewable energy alongside a 21% reduction in electricity demand since 2008. These studies provide the necessary context required for this report, which extends previous work by focusing specifically on the UK grid from 2015-2025.

Objectives This project aims to investigate how renewable energy sources and other factors influence fossil fuel power generation in the UK from 2015-2025. Exploratory visualisations are used to identify trends and seasonal patterns in the data, followed by a multiple linear regression model to quantify the relationship between each predictor variable and fossil fuel production. Specifically, the project addresses the following question: how does power generated from renewable energy sources affect fossil fuel generation, and to what extent do factors such as demand, net imports, and time of year contribute?

3 Data

Data provenance The data was retrieved from Gridwatch [6], a website that continuously monitors the UK Power Grid, by aggregating data from Elexon. The dataset was downloaded directly from the website's download page and consists of power data recorded in 5-minute intervals from 2015 to 2025. Gridwatch explicitly states that its purpose is "to provide and present the data as is", with no restrictions on access. Since the dataset is publicly available and originates from Elexon (a public body whose data is openly available), it is considered appropriate for use in this academic project.

Data description The dataset is in CSV format and consists of 1,154,090 records with no missing values. It contains variables including time-related fields (timestamp, year, month), renewable energy sources (solar, wind, hydropower, biomass), fossil fuel sources (coal, CCGT, OCGT, oil), and electricity interconnectors (French-ICT, Dutch-ICT, Irish-ICT, NEMO, NSL). All power values are in megawatts (MW).

Variables corresponding to internal electricity interconnectors, such as E-W ICT, North-South ICT were excluded from the dataset as they were irrelevant to the analysis. Additionally, the French-ICT 2, French INTELEC, and Viking ICT variables were also excluded, because they were relatively new interconnectors with smaller capacity.

Data processing The data were examined for missing values; none were found, and the column names were stripped of whitespace. Aggregate columns were created for total fossil fuel production, total renewable production, and net imports. Additional time-related columns were derived from the timestamp, including year, month, and year-month fields, to facilitate grouping and visualisation. For the regression analysis, the dataset was aggregated to daily averages to reduce short-term noise and autocorrelation in the 5-minute interval data.

4 Exploration and analysis

Figure 1 shows the monthly average renewable and fossil fuel production in the UK from 2015-2025, with trend lines fitted using least-squares regression. Renewable energy production shows a clear upward trend over the decade, while fossil fuel production shows a corresponding decline, suggesting a displacement effect over the same period. This inverse relationship motivates a deeper quantitative analysis of the other factors driving fossil fuel production.

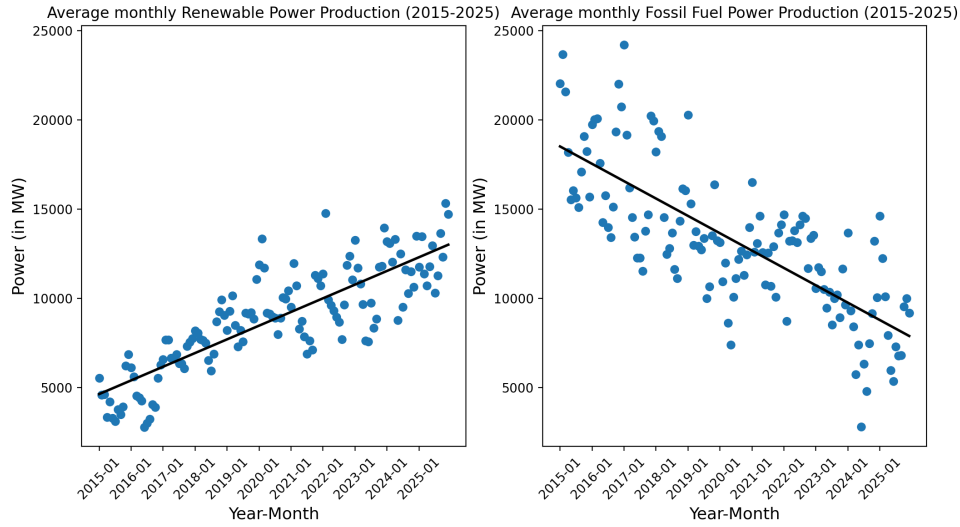


Figure 1: Monthly total renewable (left) and fossil fuel (right) energy production in the UK from 2015 to 2025, with trend lines fitted using ordinary least squares regression.

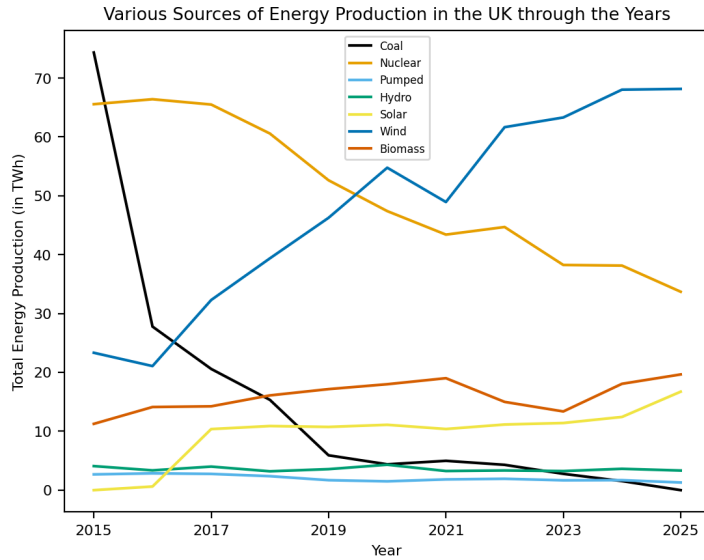


Figure 2: Total Annual Energy Production (in TWh) for various sources in the UK from 2015-2025.

achieving an R^2 of 0.968. The data were aggregated to daily averages rather than 5-minute intervals to reduce short-term noise, as discussed earlier. Net imports were included as a predictor variable under the assumption that the UK imports electricity when it is cheaper than domestic fossil fuel generation, rather than purely to fill supply gaps. The regression equation is given by Equation 1:

Figure 2 shows total annual energy production by source from 2015 to 2025. Coal declined from nearly 75 TWh in 2015 to zero by 2025, following the closure of the UK's final coal-fired power station in September 2024. Meanwhile, renewable sources experienced significant growth – wind energy in particular increased from 23 TWh in 2015 to 70 TWh in 2025, becoming the largest single source of UK electricity by 2024. Nuclear generation remained stable throughout the period, while biomass and solar also showed moderate growth.

To assess the effects of renewables, demand, net imports, and time of year on fossil fuel production, a multiple linear regression model was fitted on daily aggregated data (Table 1),

Table 1: OLS Regression Results

Dep. Variable:	fossil	R-squared:	0.968
Model:	OLS	Adj. R-squared:	0.968
Method:	Least Squares	F-statistic:	7133.
Durbin-Watson:	0.248	Condition Number:	689,000

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-6526.0225	277.768	-23.495	0.000	-7070.602	-5981.443
solar	0.0456	0.023	1.962	0.050	3.69e-05	0.091
wind	-0.9019	0.005	-166.806	0.000	-0.913	-0.891
hydro	-1.3406	0.105	-12.794	0.000	-1.546	-1.135
biomass	-0.4688	0.027	-17.429	0.000	-0.522	-0.416
demand	0.9101	0.007	137.576	0.000	0.897	0.923
net_imports	-0.8877	0.016	-55.144	0.000	-0.919	-0.856
May	-285.1394	106.272	-2.683	0.007	-493.491	-76.787
Jun	-329.1356	109.864	-2.996	0.003	-544.531	-113.740
Dec	-438.0969	78.561	-5.577	0.000	-592.119	-284.074

Notes:

(1) Standard errors assume the covariance matrix is correctly specified.

(2) The condition number is 689,000. This might indicate that there are strong multicollinearity or other numerical problems.

$$y_{fossil} = \beta_0 + \beta_1 x_{solar} + \beta_2 x_{wind} + \beta_3 x_{hydro} + \beta_4 x_{biomass} + \beta_5 x_{demand} + \beta_6 x_{net_imports} + \sum_{m=Feb}^{Dec} \beta_m D_m + \varepsilon \quad (1)$$

Here, D_m represents monthly dummy variables with January as the baseline, while ε represents the error/noise term associated with the prediction. The intercept ($\beta_0 = -6526$) represents the baseline fossil fuel production when all predictors are zero, though this has no physical interpretation as zero demand is not a realistic scenario.

Monthly dummy variables were included to control for seasonal variation, with January as the baseline. December ($p < 0.001$), May ($p = 0.007$) and June ($p = 0.003$) showed statistically significant reductions in fossil fuel production relative to January, while most other months were not significantly different from January after controlling for other predictors. Figure 3 shows the regression coefficients for the continuous predictor variables, with error bars representing their 95% confidence intervals. Each of these has a confidence interval that does not cross zero, confirming they are all statistically significant. Wind and demand show the narrowest intervals, indicating highly precise estimates, likely due to their

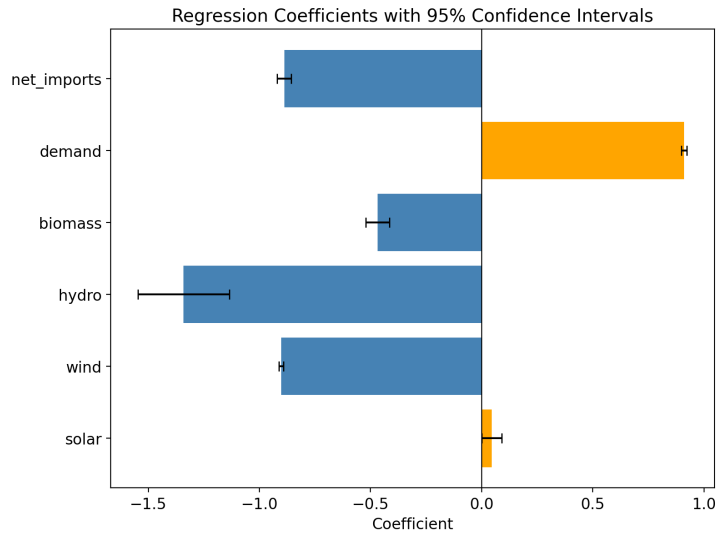


Figure 3: Continuous variables in the regression model and the coefficients with their 95% confidence intervals. Blue represents the variables with negative coefficients, while orange represents variables with positive coefficients.

large variation across the dataset. Solar has an unexpected positive coefficient ($\beta = 0.046$, $p=0.050$), with an interval just barely excluding zero, suggesting it is borderline significant and should be interpreted with caution. All other continuous predictor coefficients have p-values of less than 0.001. Wind and net imports have near-identical negative coefficients of around -0.90, while biomass and hydropower have coefficients of -0.47 and -1.34 respectively, indicating that all renewables (except solar) and net imports are negatively associated with fossil fuel production. Demand has a positive coefficient of 0.91, consistent with higher demand requiring more fossil fuel generation.

Figure 4 shows the average electricity demand and wind power production across months from 2015 to 2025, colour-coded by month. Demand exhibits a clear seasonal pattern, peaking in winter months (December-February) and reaching its lowest in summer (June-August). Wind generation similarly shows higher average output in winter months (November-February) compared to summer, and has grown significantly across all months over the decade. These seasonal patterns in both demand and wind generation motivate the inclusion of monthly dummy variables in the regression model to control for seasonal variation.

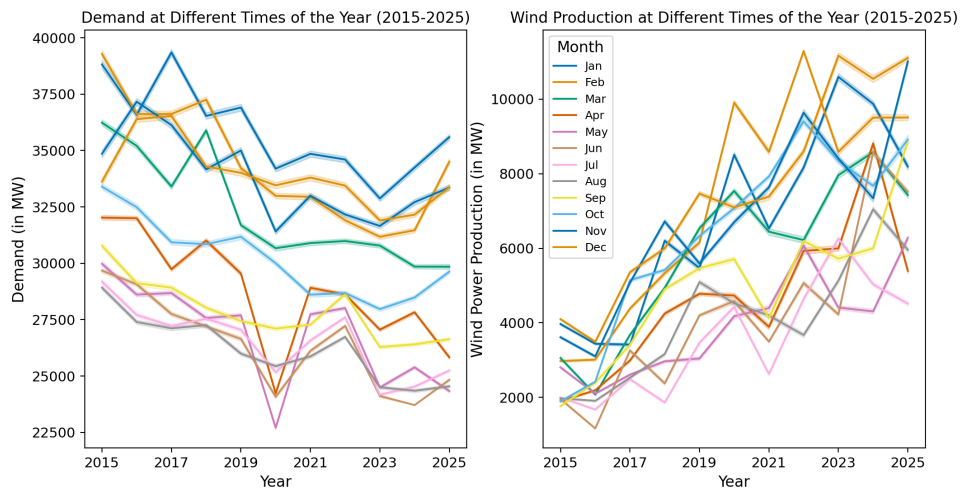


Figure 4: Demand (left) and Wind Power Production (right) at different times of the year, colour-coded by month.

5 Discussion and conclusions

Summary of findings The analysis indicates that, in general, renewable energy sources are negatively associated with fossil fuel production. Solar energy is an exception, showing a borderline positive coefficient ($\beta = 0.046$, $p=0.050$), likely reflecting multicollinearity with demand as both follow similar seasonal patterns of peaking in summer. December, May and June show statistically significant negative coefficients (Table 1), indicating lower fossil fuel production relative to January after controlling for other predictors. Demand is positively associated with fossil fuel production; the higher the demand, the more fossil fuels are needed to meet it. Net imports are negatively associated, consistent with the assumption that electricity is imported as a cheaper alternative to domestic fossil fuel generation. Hydropower has the strongest displacement effect among all renewable sources, with a coefficient of -1.34, indicating that each MW of hydropower displaces approximately 1.34 MW of fossil fuel production.

Wind energy and net imports have nearly identical coefficients of approximately -0.90, suggesting that these are effective substitutes for fossil fuels, displacing them at approximately the same rate. Wind energy in particular shows the narrowest confidence interval among all renewable sources ($[-0.913, -0.891]$),

indicating a highly precise and reliable estimate. This, combined with its statistical significance ($p < 0.001$) and high winter generation as shown in Figure 4, suggests wind energy plays a key role in compensating for higher winter demand, reducing reliance on fossil fuels during peak demand periods.

Evaluation of own work: Strengths and Limitations The model achieves an R^2 of 0.968; the predictor variables explain a large proportion of the variance in fossil fuel production. The inclusion of multiple predictors, such as renewable sources, demand, net imports, and monthly dummy variables, controls for confounding factors, allowing for a more reliable estimate of each variable's independent effect. Separating renewable sources individually rather than treating them as a single aggregate reveals differences in displacement efficiency between wind, hydro, biomass and solar. The use of daily aggregation reduces short-term noise and autocorrelation inherent in the 5-minute interval data, improving the reliability of the regression estimates. Finally, the use of confidence intervals allows for an interpretation of the statistical significance of each coefficient.

A key limitation is that regression reflects association rather than causation, and therefore, the coefficients indicate patterns in the data rather than causal effects. The low Durbin-Watson statistic of 0.248 indicates significant positive autocorrelation in the residuals, violating the OLS assumption of independent errors. The high condition number of 689,000 indicates potential multicollinearity, evidenced by the unexpected positive coefficient for solar, likely reflecting its correlation with demand. Additionally, the assumption that net imports act as a predictor variable may not always hold, as discussed in Section 4. Finally, while the model was fitted on the full dataset without a train/test split, out-of-sample performance is illustrated by using post-2022 data as a test set.

Comparison with any other related work The report is consistent with Ember's analysis [2], which reports that low-carbon renewables overtook fossil fuels as the largest source of electricity in the UK in 2024, with wind energy on course to overtake gas as the largest single source of UK electricity for the first time. Karlilar Pata and Balcilar [4] found a wind displacement coefficient of -0.55 across 36 OECD countries, which is notably lower than the -0.90 found in this report in the UK specifically. This suggests that wind energy is more effective at displacing fossil fuels in the UK than the OECD average, likely reflecting the UK's highly interconnected grid and the role of wind in the UK's power generation. Carbon Brief [5] similarly found that fossil fuel generation in the UK fell to its lowest ever share in 2023, driven by renewable expansion and reduced demand, which is consistent with the negative coefficients found for all renewable sources in this report.

This report extends previous work by quantifying the individual displacement effects of separate renewable sources (wind, solar, hydro, biomass) rather than treating renewables as a single aggregate and by including net electricity imports as an additional predictor. This reveals that net imports displace fossil fuels at a rate comparable to wind. RenewableUK [7] reports that fossil fuels fell to a record low of 26.7% in 2024, with wind outpacing gas for the third time in a quarterly period, further supporting the findings of this report.

Improvements and extensions A natural extension would be to address the autocorrelation issue by incorporating lagged fossil fuel production as an additional predictor, or by adopting a time-series modelling approach, such as ARIMA [8], which explicitly models autocorrelation structure. A formal train/test split would also provide an assessment of out-of-sample predictive performance. Separating net imports into individual interconnectors (French, Norwegian, Dutch) would allow for an analysis of whether different import sources have different displacement effects. Additionally, the model could also be used to forecast future fossil fuel production under different scenarios, such as projecting the effect of achieving the UK government's target of 95% clean power by 2030.

References

- [1] National Grid: *What we do, 2025*, Accessed: March, 2026. [Online]. Available: <https://www.nationalgrid.com/about-us/what-we-do>.
- [2] F. Mayo, 'UK low-carbon renewable power set to overtake fossil fuels for first time,' 2024, Accessed: March, 2026. [Online]. Available: <https://ember-energy.org/latest-insights/uk-renewable-power-set-to-overtake-fossil-fuels-for-first-time/>.
- [3] BBC News, *Renewables jobs not making up for north sea decline, MPs warn*, 2024, Accessed: March, 2026. [Online]. Available: <https://www.bbc.co.uk/news/articles/cz0x19rx0x2o>.
- [4] S. Karlilar Pata and M. Balcilar, 'Decarbonizing energy: Evaluating fossil fuel displacement by renewables in OECD countries,' *Environmental Science and Pollution Research*, 2024, Accessed: March, 2026. DOI: 10.1007/s11356-024-33324-8. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11096244/>.
- [5] S. Evans and V. Viisainen, *Analysis: UK electricity from fossil fuels drops to lowest level since 1957*, 2024, Accessed: March, 2026. [Online]. Available: <https://www.carbonbrief.org/analysis-uk-electricity-from-fossil-fuels-drops-to-lowest-level-since-1957/>.
- [6] Templar Consultancy Limited, *Gridwatch: GB national grid status*, Data sourced from Elexon portal and Sheffield University PV Live, Accessed: March, 2026. [Online]. Available: <https://www.gridwatch.templar.co.uk/download.php>.
- [7] RenewableUK, *First ever statistics on progress to clean power by 2030 show 74% of great britain's electricity generated by low carbon sources in 2024*, Accessed: March, 2026. [Online]. Available: <https://www.renewableuk.com/news-and-resources/press-releases/first-ever-statistics-on-progress-to-clean-power-by-2030-show-74-of-great-britain-s-electricity-generated-by-low-carbon-sources-in-2024/>.
- [8] Noble, Joshua, 'Introducing ARIMA models,' Accessed: March, 2026. [Online]. Available: <https://www.ibm.com/think/topics/arima-model>.

FDS-Project-Code

March 31, 2026

1 Imports

```
[66]: import pandas as pd
import matplotlib.pyplot as plt
import matplotlib as mpl
import seaborn as sns
import numpy as np
import statsmodels.formula.api as smf

mpl.rcParams['font.size']=8
mpl.rcParams['figure.dpi']=200
```

2 Data

```
[67]: #Read the data

data=pd.read_csv("gridwatch.csv")
```

3 Exploratory Data Analysis

```
[68]: #First 5 rows

data.head()
```

```
[68]:
```

	id	timestamp	demand	coal	nuclear	ccgt	wind	\
0	377525	2015-01-01 00:00:04	28809	9079	8049	3360	5251	
1	377526	2015-01-01 00:05:02	28645	8947	8053	3369	5254	
2	377527	2015-01-01 00:10:02	28768	8843	8052	3372	5272	
3	377528	2015-01-01 00:15:02	28917	8763	8047	3339	5303	
4	377529	2015-01-01 00:20:02	28964	8818	8051	3386	5223	

	pumped	hydro	biomass	oil	solar	ocgt	french_ict	dutch_ict	\
0	15	443	1157	0	0.0	0	582	900.0	
1	0	441	1157	0	0.0	0	586	898.0	
2	0	440	1157	0	0.0	0	586	898.0	
3	0	439	1155	0	0.0	0	586	898.0	

4	0	440	1155	0	0.0	0	586	898.0
---	---	-----	------	---	-----	---	-----	-------

	irish_ict	nemo	ns1
0	-72	0	0
1	-100	0	0
2	-100	0	0
3	-100	0	0
4	-100	0	0

[69]: *#Data description*

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1154090 entries, 0 to 1154089
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   id               1154090 non-null  int64
1   timestamp        1154090 non-null  object
2   demand           1154090 non-null  int64
3   coal             1154090 non-null  int64
4   nuclear          1154090 non-null  int64
5   ccgt             1154090 non-null  int64
6   wind             1154090 non-null  int64
7   pumped           1154090 non-null  int64
8   hydro            1154090 non-null  int64
9   biomass          1154090 non-null  int64
10  oil              1154090 non-null  int64
11  solar            1154090 non-null  float64
12  ccgt             1154090 non-null  int64
13  french_ict       1154090 non-null  int64
14  dutch_ict        1154090 non-null  float64
15  irish_ict        1154090 non-null  int64
16  nemo             1154090 non-null  int64
17  ns1              1154090 non-null  int64
dtypes: float64(2), int64(15), object(1)
memory usage: 158.5+ MB
```

[70]: *#Clean column names*

```
print(data.columns)
renamed_columns={}
for i in data.columns:
    renamed_columns[i]=i.strip()
data.rename(columns=renamed_columns,inplace=True)
print(data.columns)
```

```
Index(['id', 'timestamp', 'demand', 'coal', 'nuclear', 'ccgt', 'wind',
      'pumped', 'hydro', 'biomass', 'oil', 'solar', 'ocgt',
      'french_ict', 'dutch_ict', 'irish_ict', 'nemo', 'nsl'],
      dtype='object')
Index(['id', 'timestamp', 'demand', 'coal', 'nuclear', 'ccgt', 'wind',
      'pumped', 'hydro', 'biomass', 'oil', 'solar', 'ocgt', 'french_ict',
      'dutch_ict', 'irish_ict', 'nemo', 'nsl'],
      dtype='object')
```

[71]: *#Search for NaNs*

```
for i in data.columns:
    print(f"{i}: {data[i].isnull().sum()}")
```

```
id: 0
timestamp: 0
demand: 0
coal: 0
nuclear: 0
ccgt: 0
wind: 0
pumped: 0
hydro: 0
biomass: 0
oil: 0
solar: 0
ocgt: 0
french_ict: 0
dutch_ict: 0
irish_ict: 0
nemo: 0
nsl: 0
```

[72]: *#Convert 'timestamp' to type datetime*

```
months={1:'Jan',2:'Feb',3:'Mar',4:'Apr',5:'May',6:'Jun',7:'Jul',8:'Aug',9:
        ↪'Sep',10:'Oct',11:'Nov',12:'Dec'}
def month(x):
    return months[x]

#Make columns for year-month,month,year
data['timestamp']=pd.to_datetime(data['timestamp'])
year_month=data['timestamp'].dt.strftime('%Y-%m') #Year-month format
Year=[i.year for i in data['timestamp']]
Month=[i.month for i in data['timestamp']]
data.insert(2,'year-month',year_month)
data.insert(3,'year',Year)
data.insert(4,'month',Month)
data.insert(5,'month_name',data['month'].apply(month))
```

```

data["renewable"]=data["wind"]+data["solar"]+data["hydro"]+data["biomass"] #Add
↳renewable column
data["fossil"] = data["coal"] +data["ccgt"] +data["ocgt"] +data["oil"] #Add
↳fossil fuel column
data['net_imports']=data['french_ict']+data['irish_ict']+data['nemo']+data['nsl']
↳#Add column for net imports

data.head()

```

```

[72]:      id      timestamp year-month  year  month month_name  demand  \
0  377525  2015-01-01 00:00:04   2015-01  2015      1      Jan   28809
1  377526  2015-01-01 00:05:02   2015-01  2015      1      Jan   28645
2  377527  2015-01-01 00:10:02   2015-01  2015      1      Jan   28768
3  377528  2015-01-01 00:15:02   2015-01  2015      1      Jan   28917
4  377529  2015-01-01 00:20:02   2015-01  2015      1      Jan   28964

```

```

      coal  nuclear  ccgt  ...  solar  ocgt  french_ict  dutch_ict  irish_ict  \
0   9079      8049  3360  ...    0.0    0         582       900.0        -72
1   8947      8053  3369  ...    0.0    0         586       898.0       -100
2   8843      8052  3372  ...    0.0    0         586       898.0       -100
3   8763      8047  3339  ...    0.0    0         586       898.0       -100
4   8818      8051  3386  ...    0.0    0         586       898.0       -100

```

```

      nemo  nsl  renewable  fossil  net_imports
0      0    0      6851.0   12439         510
1      0    0      6852.0   12316         486
2      0    0      6869.0   12215         486
3      0    0      6897.0   12102         486
4      0    0      6818.0   12204         486

```

[5 rows x 25 columns]

```

[73]: #Plot total energy production for various sources
groups=data.groupby('year')

y1=groups['coal'].sum() * (5/60)/1e6 #Convert to TWh (since data is in 5
↳minute intervals)
y2=groups['nuclear'].sum() * (5/60)/1e6
y3=groups['pumped'].sum() * (5/60)/1e6
y4=groups['hydro'].sum() * (5/60)/1e6
y5=groups['solar'].sum() * (5/60)/1e6
y6=groups['wind'].sum() * (5/60)/1e6
y7=groups['biomass'].sum() * (5/60)/1e6

x=y1.index

colors = ['#000000','#E69F00','#56B4E9','#009E73','#F0E442','#0072B2',

```

```

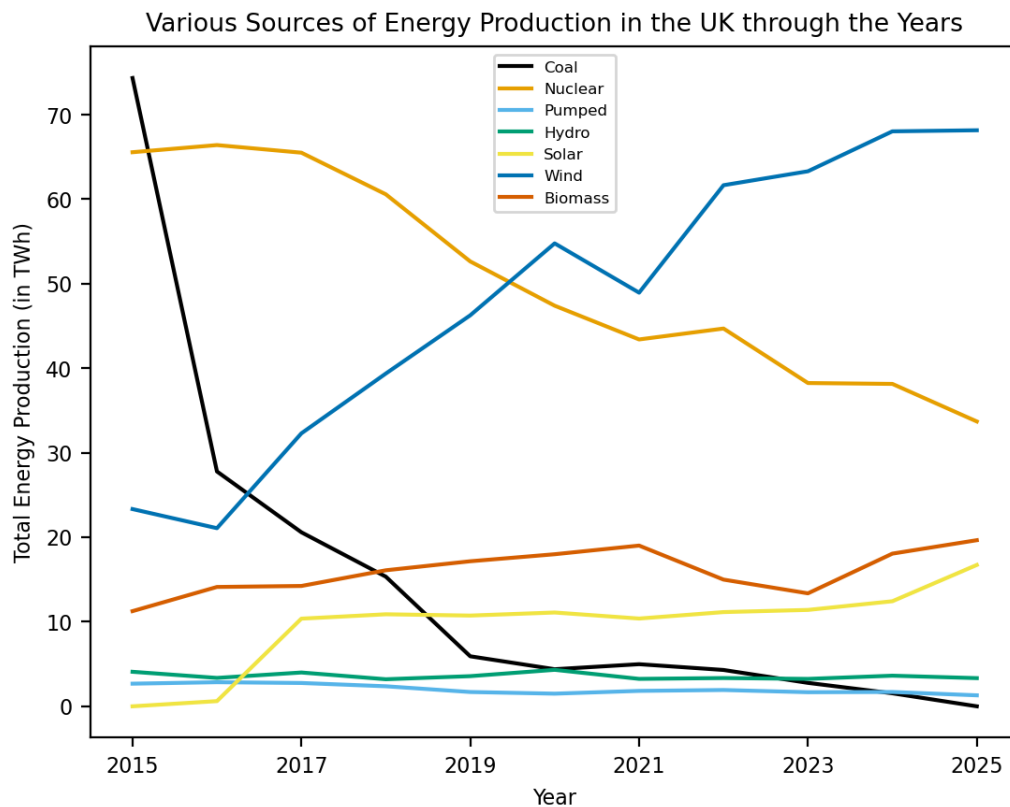
'#D55E00', '#CC79A7'] #Wong (2011) suggested colour-blind friendly
↪palette

```

```

fig, ax = plt.subplots()
ax.plot(x, y1, label='Coal', color=colors[0])
ax.plot(x, y2, label='Nuclear', color=colors[1])
ax.plot(x, y3, label='Pumped', color=colors[2])
ax.plot(x, y4, label='Hydro', color=colors[3])
ax.plot(x, y5, label='Solar', color=colors[4])
ax.plot(x, y6, label='Wind', color=colors[5])
ax.plot(x, y7, label='Biomass', color=colors[6])
plt.title("Various Sources of Energy Production in the UK through the Years")
plt.xticks(ticks=range(2015, 2026, 2))
plt.xlabel("Year")
plt.ylabel("Total Energy Production (in TWh)")
plt.legend(fontsize=6, loc='best');

```



```

[74]: month_order=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct',
                  'Nov', 'Dec']
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

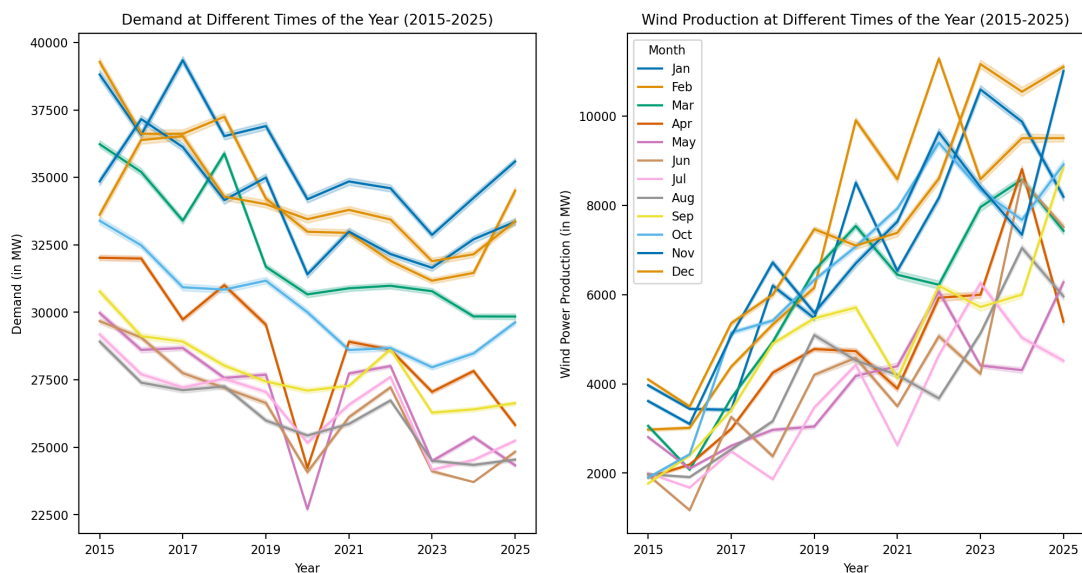
```

```

sns.
    ↳lineplot(data=data,x="year",y="demand",hue="month_name",hue_order=month_order,
              palette="colorblind",ax=axes[0],legend=False)
axes[0].set_title("Demand at Different Times of the Year (2015-2025)")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Demand (in MW)")
axes[0].set_xticks(range(2015,2026,2)) #Set x-ticks from 2015-2025 skipping
    ↳every other year

sns.lineplot(data=data,x="year",y="wind",hue="month_name",
             hue_order=month_order,palette="colorblind",ax=axes[1])
axes[1].set_title("Wind Production at Different Times of the Year (2015-2025)")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Wind Power Production (in MW)")
axes[1].legend(title='Month')
axes[1].set_xticks(range(2015,2026,2));

```

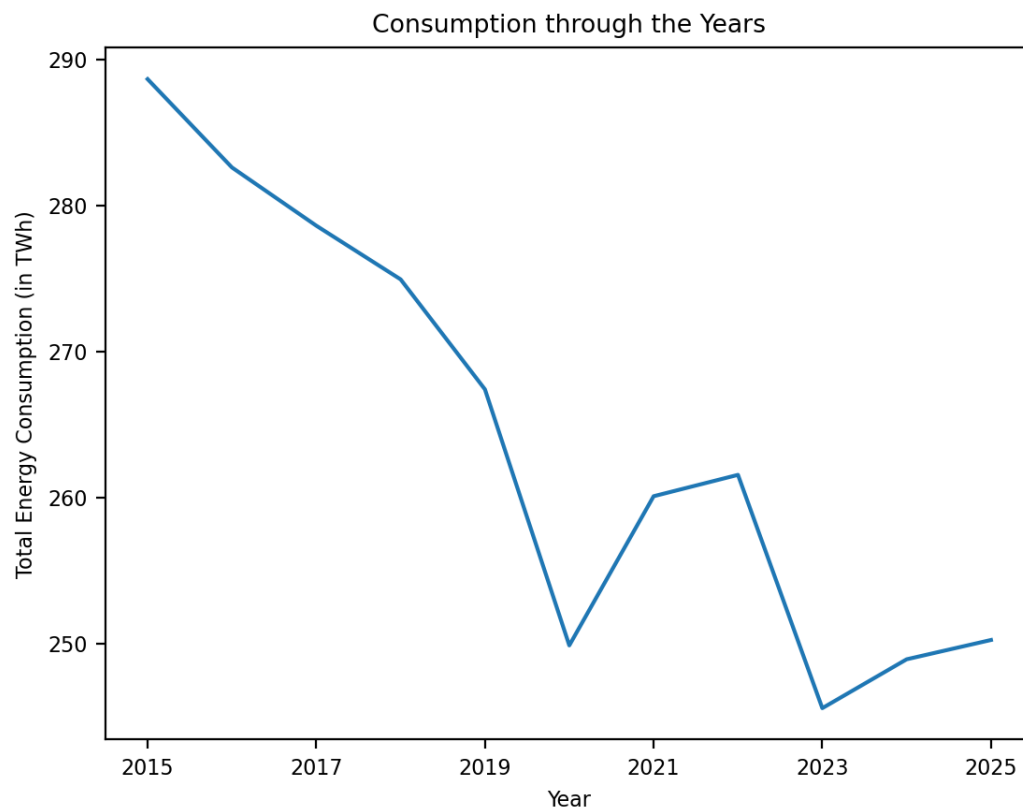


```

[75]: #Total consumption by Year

groups=data.groupby('year')
demand=groups['demand'].sum() * (5/60)/1e6
year=range(2015,2026)
plt.plot(year,demand)
plt.xlabel("Year")
plt.ylabel("Total Energy Consumption (in TWh)")
plt.title("Consumption through the Years")
plt.xticks(ticks=range(2015,2026,2));

```



4 Data Communication and Analysis

[76]: *#Plot average monthly renewable & fossil power production from 2015-2025*

```
y1=data.groupby(by='year-month')['renewable'].mean()
x1=y1.index
x1_numeric=range(len(x1))

y2=data.groupby(by='year-month')['fossil'].mean()
x2=y2.index
x2_numeric=range(len(x2))

fig=plt.figure(figsize=(12,6))
ax1,ax2=fig.subplots(1,2,sharey=True)

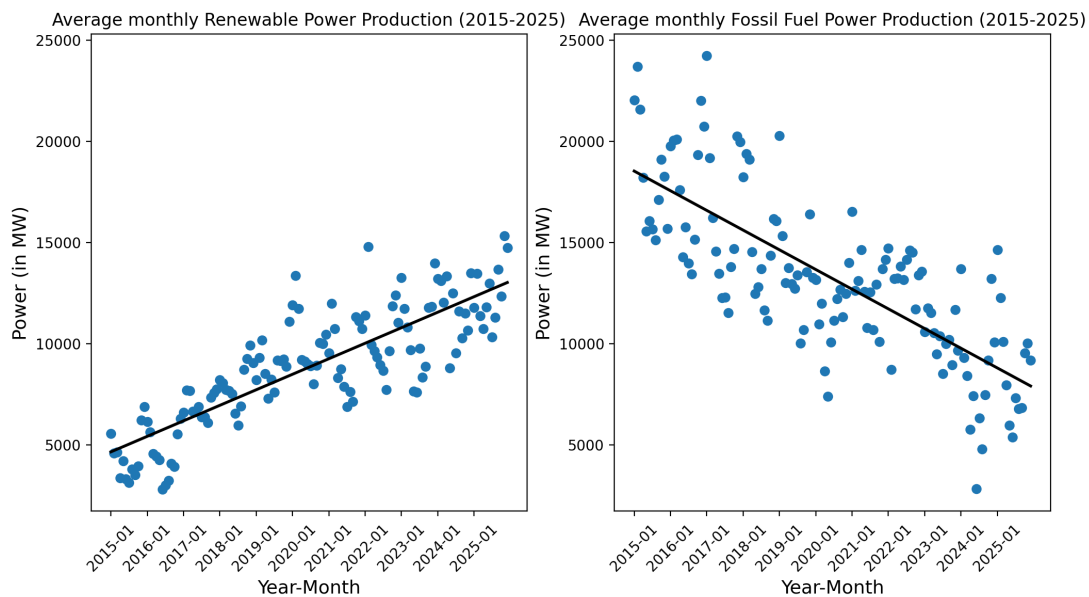
ax1.scatter(x=x1,y=y1)
z1=np.polyfit(x1_numeric,y1,1) #Returns least squares polynomial fit
```

```

ax1.plot(x1,np.poly1d(z1)(x1_numeric),color='black',linewidth=2) #Line of best fit
ax1.set_title("Average monthly Renewable Power Production (2015-2025)",fontsize=12)
ax1.set_xlabel("Year-Month",fontsize=13)
ax1.set_ylabel("Power (in MW)",fontsize=13)
ax1.set_xticks(x1[::12])
ax1.tick_params(axis='x',rotation=45)
ax1.tick_params(axis='both',labelsize=10)

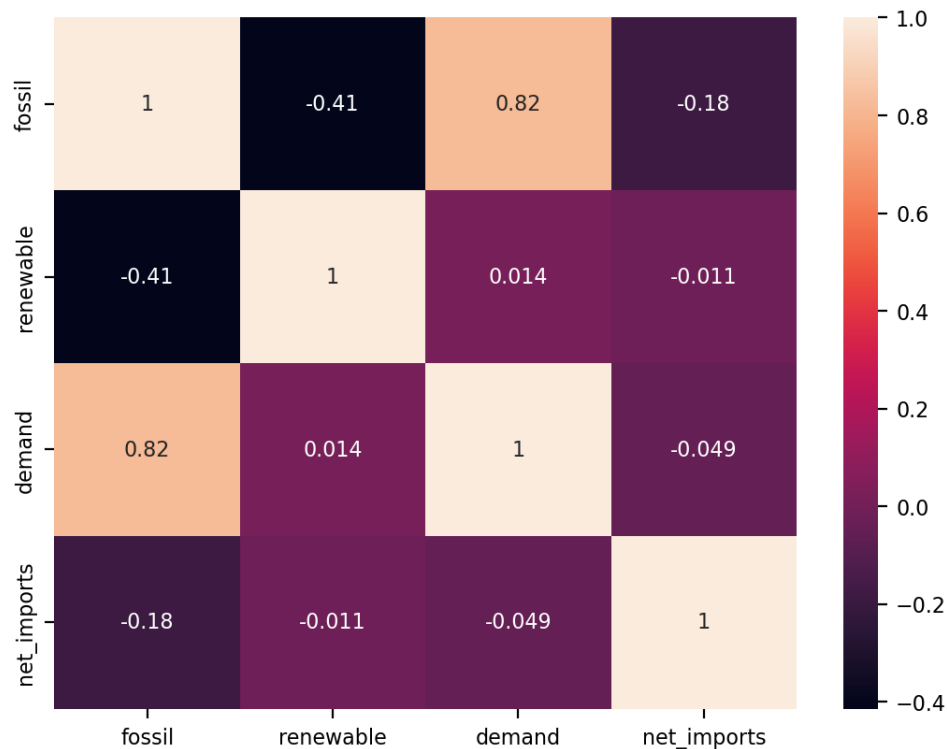
ax2.scatter(x=x2,y=y2)
z2=np.polyfit(x2_numeric,y2,1)
ax2.plot(x2,np.poly1d(z2)(x2_numeric),color='black',linewidth=2)
ax2.set_title("Average monthly Fossil Fuel Power Production (2015-2025)",fontsize=12)
ax2.set_xlabel("Year-Month",fontsize=13)
ax2.set_ylabel("Power (in MW)",fontsize=13)
ax2.set_xticks(x2[::12])
ax2.tick_params(labelleft=True)
ax2.tick_params(axis='x',rotation=45)
ax2.tick_params(axis='both',labelsize=10)

```



[77]: *#Correlation matrix for all the variables concerned*

```
sns.heatmap(data[["fossil","renewable","demand","net_imports"]].
↪corr(),annot=True);
```



[78]: *#Multiple regression model using OLS*

```
daily_data=data.resample('D',on='timestamp').mean(numeric_only=True) #Aggregate
↪daily
daily_data.insert(3,'month_name',daily_data.index.strftime('%b'))
daily_data['month_name']=pd.
↪Categorical(daily_data['month_name'],categories=month_order,ordered=True)
↪#Include monthly dummies with January as the baseline at 0
model_2=smf.ols(formula='fossil ~ solar + wind + hydro + biomass + demand +
↪net_imports + month_name', data=daily_data).fit()
model_2.summary()
```

[78]:

Dep. Variable:	fossil	R-squared:	0.968
Model:	OLS	Adj. R-squared:	0.968
Method:	Least Squares	F-statistic:	7133.
Date:	Tue, 31 Mar 2026	Prob (F-statistic):	0.00
Time:	00:41:56	Log-Likelihood:	-33444.
No. Observations:	4016	AIC:	6.692e+04
Df Residuals:	3998	BIC:	6.704e+04
Df Model:	17		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-6526.0225	277.768	-23.495	0.000	-7070.602	-5981.443
month_name[T.Feb]	-125.4607	79.233	-1.583	0.113	-280.801	29.880
month_name[T.Mar]	112.8951	81.157	1.391	0.164	-46.218	272.009
month_name[T.Apr]	-180.0350	96.263	-1.870	0.062	-368.765	8.695
month_name[T.May]	-285.1394	106.272	-2.683	0.007	-493.491	-76.787
month_name[T.Jun]	-329.1356	109.864	-2.996	0.003	-544.531	-113.740
month_name[T.Jul]	-243.4964	108.377	-2.247	0.025	-455.977	-31.016
month_name[T.Aug]	-100.0147	107.156	-0.933	0.351	-310.100	110.070
month_name[T.Sep]	-143.7492	100.080	-1.436	0.151	-339.963	52.464
month_name[T.Oct]	18.9436	87.367	0.217	0.828	-152.344	190.231
month_name[T.Nov]	-143.4783	80.073	-1.792	0.073	-300.467	13.510
month_name[T.Dec]	-438.0969	78.561	-5.577	0.000	-592.119	-284.074
solar	0.0456	0.023	1.962	0.050	3.69e-05	0.091
wind	-0.9019	0.005	-166.806	0.000	-0.913	-0.891
hydro	-1.3406	0.105	-12.794	0.000	-1.546	-1.135
biomass	-0.4688	0.027	-17.429	0.000	-0.522	-0.416
demand	0.9101	0.007	137.576	0.000	0.897	0.923
net_imports	-0.8877	0.016	-55.144	0.000	-0.919	-0.856
Omnibus:	27.111	Durbin-Watson:	0.248			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	27.025			
Skew:	0.187	Prob(JB):	1.35e-06			
Kurtosis:	2.852	Cond. No.	6.89e+05			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 6.89e+05. This might indicate that there are strong multicollinearity or other numerical problems.

[79]: *#Plot the regression coefficients continuous variables*

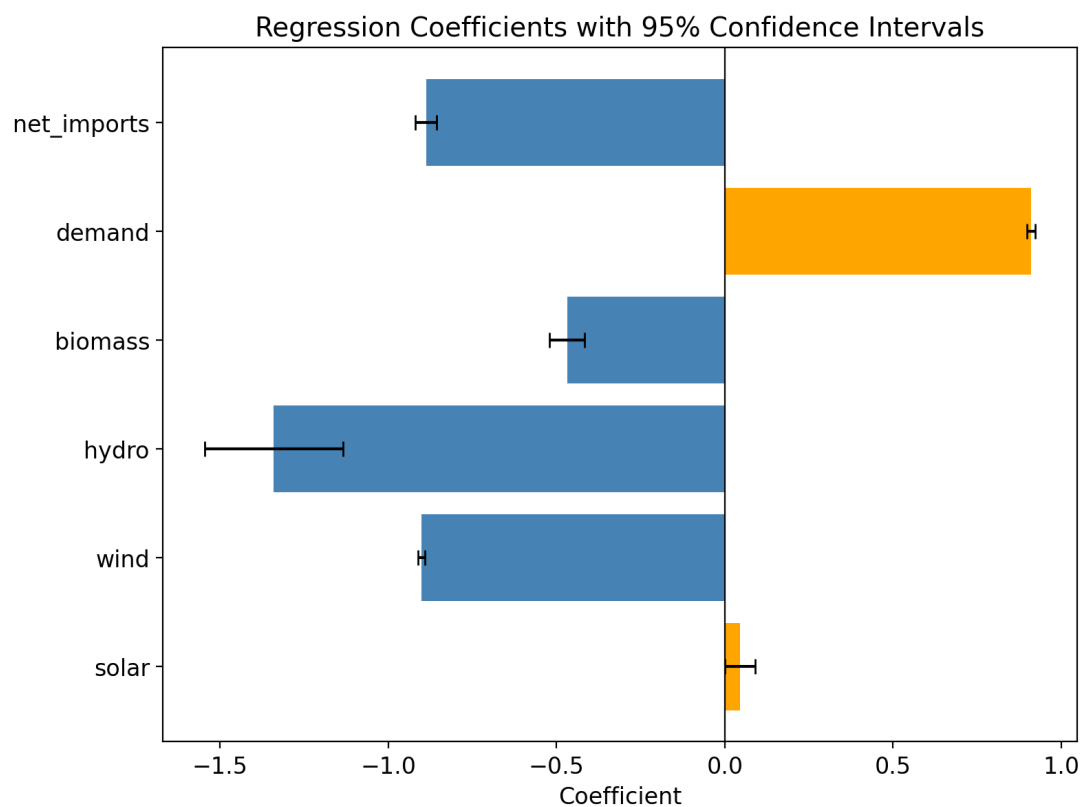
```
fig,ax=plt.subplots(figsize=(8,6))
continuous=['solar','wind','hydro','biomass','demand','net_imports']
coefs=model_2.params[continuous]
conf=model_2.conf_int().loc[continuous] #Confidence intervals

ax.barh(continuous,coefs,xerr=[coefs-conf[0],conf[1]-coefs],capsize=4,
```

```

        color=['orange' if c > 0 else 'steelblue' for c in coefs]) #Blue when
        ↪coefficient<0, orange otherwise
ax.axvline(x=0,color='black',linewidth=0.8)
ax.set_xlabel('Coefficient',fontsize=12)
ax.set_title('Regression Coefficients with 95% Confidence
        ↪Intervals',fontsize=14)
ax.tick_params(axis='both',labelsize=12)
plt.tight_layout()
plt.show()

```



```

[80]: #Plot actual vs predicted data to see how well the model fits

test_data=daily_data[daily_data['year']>2022] #Test data with year>2022 for
        ↪70-30 split
predictions=model_2.predict(test_data)
actual=test_data['fossil']

plt.scatter(actual,predictions)
plt.xlabel("Actual Power Production (in MW)")
plt.ylabel("Predicted Production (in MW)")

```

```
plt.title("Actual vs Predicted Fossil Fuel Production (Daily)")  
plt.plot([actual.min(),actual.max()], [actual.min(),actual.max()],color='black')  
plt.show()
```

